Clean Architecture & Object-Oriented Design Handbook

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1. The Core Philosophy of Clean Architecture

Clean Architecture organizes software systems so that business logic remains completely independent of frameworks, UI libraries, databases, and external I/O protocols. Key rules include:

" The Dependency Rule: Source code dependencies must only point inwards toward high-level policies.

" Entities: Business rules that encapsulate enterprise-wide critical business operations.

" Use Cases: Application-specific rules that orchestrate data flow between entities and interfaces.

2. SOLID Design Principles in Practice

Uncle Bob formalized 5 foundational object-oriented design principles:

" Single Responsibility Principle (SRP): A class or module should have only one reason to change.

" Open/Closed Principle (OCP): Software artifacts should be open for extension, closed for modification.

" Liskov Substitution Principle (LSP): Subtypes must be substitutable for their base types.

" Interface Segregation Principle (ISP): Clients should never be forced to depend on unused methods.

" Dependency Inversion Principle (DIP): High-level modules must not depend on low-level details.